## Your grade: 96%

Your latest: **96%** • Your highest: **96%** • To pass you need at least 80%. We keep your highest score.

Next item →

# Assessment 10

**1.** Which of the following qu... ged after applying the CNOT gate? **1 / 1 point**

- [x] $|00\rangle$

Correct
Correct! $|00\rangle$ remains unchanged after applying the CNOT gate.

- [ ] $|10\rangle$
- [ ] $|11\rangle$
- [x] $|01\rangle$

Correct
Correct! $|01\rangle$ remains unchanged after applying the CNOT gate.

**Review Learning Objectives**

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice    Let's chat

**Assignment details**

**Due**
Dec 23, 11:59 PM IST

**Attempts**
Unlimited

Retry

**Submitted**
Dec 11, 1:39 AM IST

**Your grade**
To pass you need at least 80%. We keep your highest score.

**96%**

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**2.** Which quantum gate is also known as the controlled-NOT gate? **1 / 1 point**

- (●) CNOT gate
- ( ) Swap gate
- ( ) Pauli-X gate
- ( ) Hadamard gate

Correct
Correct! The CNOT gate, or controlled-NOT gate, is a fundamental two-qubit gate in quantum computing.

**3.** In a quantum circuit, what is the effect of applying the Hadamard gate to a qubit initially in the $|0\rangle$ state? **1 / 1 point**

- ( ) The qubit remains in the $|0\rangle$ state
- (●) The qubit enters a superposition state $\frac{|0\rangle+|1\rangle}{\sqrt{2}}$
- ( ) The qubit enters the $|1\rangle$ state
- ( ) The qubit enters a mixed state of $|0\rangle$ and $|1\rangle$

Correct
Correct! Applying the Hadamard gate to the $|0\rangle$ state results in a superposition state $\frac{|0\rangle+|1\rangle}{\sqrt{2}}$.

**4.** Which of the following are applications of the Hadamard gate in quantum computing? **0.8 / 1 point**

- [ ] Flipping the state of a qubit
- [ ] Entangling qubits
- [x] Creating superposition in qubits

Correct
Correct! The Hadamard gate is used to create superposition in qubits.

- [ ] Measuring the state of qubits
- [ ] Preparing qubits for interference

You didn't select all the correct answers

**5.** What is the significance of quantum entanglement in quantum computing? **1 / 1 point**

- ( ) It allows quantum computers to operate at zero temperature.
- ( ) It enables quantum bits to exist in both 0 and 1 states simultaneously.
- ( ) It is necessary for quantum bits to perform error correction.
- (●) It allows qubits to be correlated in a way that classical bits cannot be.

Correct
Correct! Quantum entanglement enables correlations that are impossible with classical bits, which is fundamental to quantum computing.